

OLO KEYSTROKE INJECTOR

Custom Payload Creation & Injection Guide

1. Introduction to Keystroke Injection

The OLO Keystroke Injector acts as a hardware emulated Human Interface Device (HID) keyboard. When connected to a host computer over Bluetooth Low Energy (BLE), the OLO device can send pre-configured keystrokes and macros automatically. Keystroke injection is a powerful technique utilized during security audits, penetration testing, and general task automation.

Because the OLO device mimics a standard user typing on a physical keyboard, the operating system accepts and processes its commands without requiring administrator privileges or specialized API permissions. Therefore, the success of a payload depends on two critical factors: focus control (the correct window must be in focus) and OS timing (matching key delay speeds with target OS responsiveness).

2. Scripting Formats and Syntaxes

2.1 Windows OS Run Mode (cmd: Prefix)

The most effective way to launch applications, navigate to web URLs, or run shell scripts on a Windows PC is by utilizing the prefix 'cmd:'. When the OLO firmware detects this prefix, it automatically executes the following automated keyboard sequence:

- Presses Win + R to open the Windows 'Run' dialog box.
- Waits 600 milliseconds to guarantee the Run window is fully rendered and focused.
- Types the text following the 'cmd:' prefix character-by-character with a safe 15ms typing interval.
- Presses the Enter key (KEY_RETURN) to execute the command.

This mechanism is extremely robust because it circumvents manual terminal navigation. Examples:

Payload String	Resulting Action on Target System
cmd:notepad	Opens the default Windows Notepad editor.
cmd:calc	Launches the Windows Calculator utility.
cmd:https://github.com	Opens GitHub inside the default web browser.
cmd:cmd /k ipconfig	Launches Command Prompt and prints active network adapters.

2.2 Plain Text Mode (Direct Typing)

If the payload string does not start with the 'cmd:' prefix, the OLO device behaves as a straight text injector. It types the exact string character-by-character into whatever text input or terminal window is currently active. Each keystroke is sent with a 15ms key delay to ensure the host operating system registers all letters accurately.

Examples:

- Payload: 'Hello World' -> Action: Types 'Hello World' directly into the cursor location.
- Payload: 'powershell.exe' -> Action: Types the string 'powershell.exe' (useful if a command line is already open).

3. Creating and Deploying Custom Payloads

OLO provides a dual deployment model, allowing users to trigger payloads wired over USB or wirelessly over Wi-Fi.

3.1 Wireless Dashboard Deployment (Wi-Fi)

1. Ensure the Wi-Fi radio is toggled ON on the device. Double-tap the touch button on the WiFi clock face screen to toggle.
2. Connect your computer or mobile phone to the same local network.
3. Open your browser and navigate to <http://olokeys.local/>
4. Authenticate by typing the administrator password.
5. Under the 'Keystroke Injection' tab, locate the 'Dynamic Key Text Payload' input box.
6. Write your payload (e.g., cmd:mspaint).
7. Click Trigger Keystroke to run immediately, or click Save Offline Payload to store the string on the device's physical non-volatile storage (NVS) for offline triggers.

3.2 Wired Companion Setup (Web Serial)

1. Connect the OLO device to your computer via a USB-C data cable.
2. Open the index.html setup utility locally in Google Chrome or Edge.
3. Click 'Connect to ESP32' and select the matching COM/Serial port.
4. Navigate to the payload injection section to instantly type custom commands or configure NVS storage profiles.

3.3 Running Saved Payloads Offline

To run the custom payload stored in the device's NVS without any network connection:

1. Touch the sensor once to wake the device, then double-tap on the clock face to open the menu.
2. Single-tap to scroll down until 'Custom Payload' is highlighted.

3. Double-tap to enter the category.
4. Single-tap to highlight 'Run Payload'.
5. Single-tap to trigger execution. The device will start the typing animation and inject the stored keystroke sequence.

4. Safety, Security and Best Practices

Keystroke injection executes exactly what is configured at very high speeds. To avoid unintended issues:

- Focus Validation: Always make sure the correct focus target window is active on the host machine before executing a payload. Injecting commands into a random browser window or terminal can lead to syntax corruption or unintended data entry.
- Timing Tolerances: If a host machine is laggy, commands sent via Win+R may type before the dialog box opens. Keep command names short or build in sleep/wait chains if calling command lines.
- Network Security: Keep your hosted dashboard password robust and disable Wi-Fi/Bluetooth status clocks when they are not in use to conserve battery and reduce wireless visibility.

5. Study Guides & Video Resources

To deepen your understanding of USB Rubber Ducky scripting, DuckyScript 3.0, and HID injection hardware engineering, use the following curated learning resources, web links, and YouTube video tutorials:

5.1 Official Documentation & Coding Environments

- **Hak5 USB Rubber Ducky Official Documentation:**
<https://docs.hak5.org/usb-rubber-ducky/>
(The primary source for scripting, payload syntax, and hardware initialization guides.)
- **Hak5 PayloadStudio Coding Environment:**
<https://payloadstudio.hak5.org/>
(A web-based IDE for writing, compiling, and debugging DuckyScript keystroke injections.)
- **Hak5 Community Payload Repository:**
<https://github.com/hak5/usbrubberducky-payloads>
(Explore real-world security audit, diagnostics, and system automation scripts written by specialists.)

5.2 Video Tutorials & Demonstrations (YouTube)

- **Hak5 Darren Kitchen Channel - Official DuckyScript Lessons:**
<https://www.youtube.com/user/Hak5Darren>
(High-quality, step-by-step video courses on HID attacks and security operations.)

- **Lesson 1: How to Inject Keystrokes in 25 Milliseconds:**

https://www.youtube.com/watch?v=e_p2fW1l848

(Deep dive into timing optimization and fast key transmission parameters.)

- **Lesson 2: Payload Triggering Tricks & Stealth Tactics:**

<https://www.youtube.com/watch?v=J3mC5b4Lz9w>

(Advanced triggers, timing considerations, and bypass techniques.)

- **Lesson 3: Staged vs. Non-Staged Payload Architectures:**

https://www.youtube.com/watch?v=k_jK4G_3M8g

(Learn how to split payloads into lightweight loaders and robust backend commands.)